Total Printed Pages: 04

Roll No. : 0108IT241015

Paper Code: MAB-101
Examination –Nov.- 2024
B.Tech. I Sem : Common the all branch.
Subject name: Linear Algebra and Calculus

Time : 3 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any. Word limit be observed as follows:
Part (a)- Objective type/Fill in the blanks/True & False
Part (b) – Three/four sentences based answer question (Maximum 50words)
Part (c) – Short Descriptive Ans./Numerical/ algorithm /programs based Question. (Maximum 100 words)
Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)
Word limit NOT to be followed for diagram, numerical, derivation.

		Marks	BL	CO
Q.1	(a) (I) The minimum value of $x^2 + 250/x$ is (i) 75 (ii) 25 (iii) 50 (iv) 0 (II) If f(x) be a continuous function and its derivative up to n^{th} order and higher order exists. Then Maclaurin's theorem f(x) = (i) $1+x+\frac{x^2}{!2}+\frac{x^3}{!3}+\ldots\ldots\ldots\ldots$ (ii) $f(a)+xf'(a)+\frac{x^2}{!2}f''(a)+\ldots\ldots\ldots\ldots$ (iii) $f(o)+xf'(o)+\frac{x^2}{!2}f''(o)+\ldots\ldots\ldots\ldots$ (iv) $f(a)+(x-a)f'(a)+\frac{(x-a)^2}{!2}f''(a)+\ldots\ldots\ldots\ldots$	02	2	1
	(b) If $y=f(x)$ then What is the Cartesian formula for radius of curvature.	02	1	1
	(c) Using Macraurin's Theorem to prove that $\log(1+e^x)=\log 2+\frac{x}{2}+\frac{x^2}{8}-\frac{x^4}{192}+\ldots\ldots\ldots\ldots$	03	3	1
	OR			
	Find the maximum value of the function $x^3y^2(1-x-y)$	03	2	1
	(d) Find the radius of curvature of the following curve at the point $(a/4, a/4)$ on the curve $\sqrt{x}+\sqrt{y}=\sqrt{a}$	05	2	1
	OR			

Show that the function $\sin 3x - 3\sin x$ is minimum when $x = \pi/2$ and maximum when $x = 3\pi/2$

05 | 2 | 1

Q.2 (a) (I) Euler's Theorem on Homogeneous function if z is a Homogeneous function of x,y of order n then.

(i) $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = nz$ (ii) $x^2\frac{\partial z}{\partial y} + y^2\frac{\partial z}{\partial x} = nz$

(iii) $x\frac{\partial z}{\partial y} + y\frac{\partial z}{\partial x} = nz$ (iv) $y^2\frac{\partial z}{\partial x} + x^2\frac{\partial z}{\partial y} = nz$

02 | 1 | 2

(II) The function $f(x) = \frac{x^3 + y^3}{x^2 + y^2}$ is a Homogeneous function of degree.

(i) 2 (ii) 3 (iii) 1 (iv) 0

(b) If $u = e^{xyz}$, Find the value of $\frac{\partial^2 u}{\partial x\,\partial y}$

02 | 5 | 2

(c) If $u = \frac{x^4 + y^4}{x + y}$ Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3\tan u$

03 | 3 | 2

OR

If $u = e^{xyz}$ find the value of $\frac{\partial^3 u}{\partial x \partial y \partial z}$.

03 | 5 | 2

(d) If $u = \tan^{-1}\left\{\left(x^3 + y^3\right)/(x - y)\right\}$

Prove that

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2\cos 3u \sin u$$

05 | 3 | 2

OR

(i) What are the Mathematical Errors.

(ii) Evaluate $\sqrt{99}$ approximately.

05 | 1 | 2

Q.3 (a) (I) $\lim_{n\to\infty}\sum_{r=1}^{n-1}\frac{1}{n}\sqrt{\frac{n+r}{n-r}} =$

02 | 2 | 3

(i) $\int_0^1 \frac{1-x}{\sqrt{1-x^2}}dx$ (ii) $\int_0^\pi \frac{1-x}{\sqrt{1-x^2}}dx$

(ii) $\int_0^1 \frac{1+x}{\sqrt{1-x^2}}dx$ (iv) $\int_0^1 \frac{1-x}{\sqrt{1+x^2}}dx$

(II) r^{th} term of $\lim_{n\to\infty}\left[\frac{n}{n^2+1}+\frac{n}{n^2+2^2}+\frac{n}{n^2+3^2}+\frac{n}{n^2+4^2}+\ldots\ldots\frac{1}{2n}\right]$

(i) $\frac{n}{n^2+r}$ (ii) $\frac{n^2}{n^2+r^2}$ (iii) $\frac{n}{n^2-r^2}$ (iv) None of them

02	5	3
03	5	3

(b) Evaluate $\int_0^a\int_0^b xy\,dx\,dy$

(c) Evaluate $\int_0^2\int_0^x\int_0^{x+y} e^x(y+2z)\,dx\,dy\,dz$

OR

03	5	3

Find the value of

$$\lim_{n\to\infty}\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)^{1/2}\left(1+\frac{3}{n}\right)^{1/3}\ldots\ldots\ldots\ldots\left(1+\frac{n}{n}\right)^{1/n}\right]$$

(d) (i) Evaluate $\int_0^1\int_0^1 \frac{dx\,dy}{\sqrt{1-x^2}\sqrt{1-y^2}}$

05	5	3

(ii) Evaluate $\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} xy\,dy\,dx$

OR

05	4	3

Find the area of the region bounded by the curves $y^2=4-x$ and $y^2=x$

Q.4 (a) (I) The square matrix A is singular if $|A|$ is equal to

(i) 0 (ii) 1 (iii) -1 (iv) None of them.

02	1	4

(II) If λ is a characteristics root of a matrix A, then a characteristic root of A^{-1} is

(i) $1/\lambda$ (ii) λ (iii) λ^2 (iv) $1/\lambda^2$

(b) State the Cayley-Hamilton theorem for the square matrix.

02	1	4

(c) For what value of λ the set of equations fail to have unique solution

$3x-y+\lambda z=1,\quad 2x+y+z=2,\quad x+2y-\lambda z=1$

03		4

OR

Define adjoint of a matrix and find the inverse of A

$$A=\begin{bmatrix}1 & 0 & 2\\ 0 & 2 & 1\\ 2 & 0 & 3\end{bmatrix}$$

03	3	4

(d) If $A=\begin{bmatrix}0 & 1 & 2\\ 2 & -3 & 0\\ 1 & 1 & -1\end{bmatrix}$ is matrix then show that Cayley Hamilton theorem is satisfied by A and also find its inverse. | 05 | 3 | 4 |

OR

Test the consistency and solve the equations $5x+3y+7z=4,\quad 3x+26y+2z=9,\quad 7x+2y+10z=5$ | 05 | 3 | 4 |

Q.5 (a) (I) An Isolated vertex has the degree | 02 | 1 | 5 |

(i) 1 (ii) 2 (iii) 0 (iv) 3

(II) In a graph the number of odd degree vertices is always.

(i) Odd (ii) even (iii) Zero (iv) none of them

(b) State Associative law in a Boolean algebra. | 02 | 1 | 5 |

(c) Define isomorphic, Euler and Hamiltonian Graphs. | 03 | 1 | 5 |

OR

State and prove the Distributive Law in Boolean Algebra. | 03 | 2 | 5 |

(d) Show that in a Graph (Connected or Disconnected) has exactly two vertices of odd degree there must be a path joining these two vertices. | 05 | 2 | 5 |

OR

(i) Prove that $(p\vee q)\wedge(\sim p)\wedge(\sim q)$ is a contradiction by using Truth table. | 05 | 3 | 5 |

(ii) Show that

$a*b+a*b'+a'*b+a'*b'=1 \quad a,b\in B$

Where B is a Boolean algebra $(B,+,*,')$

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5-Evaluating, 6-Creating

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